Lesson plan · for the teacher

Pilot, place and test

IoT · Level 4 · Lesson 2

Time
40 min

Materials
One thermometer (a cheap room or aquarium thermometer works; plastic or digital is safest); A watch or phone to time each reading; Paper for a class table: place, time, reading, notes; Optional: a CO_2 sensor, or a microcontroller with a temperature sensor

Learning goals

- Pilot one sensor node before building many, with a written test.
- Place sensors where they measure what you mean.
- Set alert thresholds that balance false alarms against missed problems.

Activity: Placement hunt

1. Warm-up (5 min): Tell the story of ten CO_2 sensors built before testing one, all placed by open windows. Ask: what should the team have done first?
2. Predict (5 min): Sketch the room on the board and mark five places: window, door, middle, back, sunny spot. Groups predict which will read warmest and coolest, and why.
3. Measure (15 min): One group at a time places the thermometer, waits 3 minutes for it to settle, and calls out the place, the time and the reading for the class to record.
4. Compare (10 min): Fill the class table on the board. Groups compare it with their predictions and explain the differences: sun, draughts, people, heaters.
5. Decide (5 min): The class chooses the best spot for a permanent sensor and writes one reason. Exit ticket: one good place, one bad place.

Before class, check how long your thermometer takes to settle and choose five safe spots. Glass thermometers break: use a plastic or digital one if you can, and keep it away from heaters and sockets.

Discussion

- Which spot gave the strangest reading, and why?
- What would a permanent sensor in that spot have told the school?
- For your project, which is worse: a false alarm or a miss?

Exit ticket

- Name one good place and one bad place for a classroom sensor, with a reason for each.

Try it at school tomorrow
Placement hunt: With one thermometer (or a CO_2 sensor if you have one), pupils predict, then take readings in five places in the classroom: by the window, by the door, in the middle, at the back and in the sun. They record each reading with its place and time, compare them, and choose the best spot for a permanent sensor, with a reason.

Practice questions · for you

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1 **The tank alert kept going off for nothing, so a colleague moved the threshold until it never goes off. What do you suggest?**

- ◯ Put the old threshold back: false alarms are the price of safety
- ◯ Leave it there: no alerts means no wasted trips up the tower
- ◯ Test a few thresholds on the pilot's data, then choose

2 **Match each sensor to a good place for it.**

Classroom CO_2 sensor	• • • •	On the lid, pointing down, away from the inlet
Tank level sensor	• • • •	Open ground, away from roofs and trees
Soil moisture probe	• • • •	In shade, in a ventilated box
Air temperature sensor	• • • •	Seated head height, 1–2 m from people
Rain gauge	• • • •	At root depth, in the bed, not the path

3 **In your pilot, a DS18B20 temperature sensor reads 85 °C for a moment each time it powers up, then normal values. It is probably working normally.**

◯ True ◯ False

4 **Which belong in the written test for a pilot CO_2 sensor?**

- ☐ It shows a green light when switched on
- ☐ It reads the same as the sensor in the next classroom
- ☐ Outdoors, it reads close to 420 ppm
- ☐ It sends readings all week with no gaps
- ☐ It rises as the class fills and falls when windows open

5 **Your pilot sensor ran for 20 school days beside a trusted CO_2 meter. An alert at 1,000 ppm on the pilot would have gone off on 6 days, but on 2 of them the trusted meter stayed below 1,000. On 3 other days the trusted meter went above 1,000 and the pilot gave no alert.**

That threshold gives _____ false alarms and _____ misses. Lowering it would mean fewer misses but more _____.

Word bank: false alarms · misses · 3 · 2 · 6

✎ **Where in your school would it be easy to place a sensor badly? What would it measure instead of what you mean?**

Check your answers

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1 **Test a few thresholds on the pilot's data, then choose**

An alert that never goes off misses every real problem, and for a tank a miss means a day without water: far worse than a wasted trip up the tower.

2 **Classroom CO_2 sensor — Seated head height, 1–2 m from people**
Tank level sensor — On the lid, pointing down, away from the inlet
Soil moisture probe — At root depth, in the bed, not the path
Air temperature sensor — In shade, in a ventilated box
Rain gauge — Open ground, away from roofs and trees

Each place makes the sensor measure what you mean: the air pupils breathe, the water level not the splashes, the roots' soil not the trodden path, the air not the sun, the rain not the drips.

3 **True**

That's a known glitch: a DS18B20 reports 85 °C, its power-up value, until its first real reading is ready.

4 **Outdoors, it reads close to 420 ppm; It sends readings all week with no gaps; It rises as the class fills and falls when windows open**

A good test checks the readings against things you know: outdoor air is about 420 ppm, a full class raises CO_2, and fresh air lowers it; and it checks the data keeps coming.

5 **That threshold gives 2 false alarms and 3 misses. Lowering it would mean fewer misses but more false alarms.**

False alarms are alerts when the air was fine: 2 of the 6.